Four supports retain two conjugate coefficient choices

Pattern A	Pattern B	Pattern C	Pattern D
$u = +3w$	$u = +3w$	$u = +w$	$u = +w$
$u = -3w$	$u = -3w$	$u = -w$	$u = -w$

All eight normalized tuples have $r = 6$ and $a_5 = 1$
Necessary system survivors
Arithmetic solutions are not constructed